

Generative AI vs Agentic AI

English Notes

Agentic AI using LangGraph (2026) — Playlist Notes

Video Introduction

New playlist: **Agentic AI using LangGraph (2026)**

First topic: Generative AI vs Agentic AI

Goal: Understand the evolution from GenAI to Agentic AI through a practical scenario.

1 What is Generative AI?

Definition: A class of AI models that can create new content (text, images, audio, code, video) resembling human-created data.

1.1 Key Evolution Timeline

- Emerged ~3 years ago (around 2023)
- Transformative technology that changed the world

1.2 Popular GenAI Products

Category	Examples
Chatbots	ChatGPT, Gemini, Claude, Grok
Image Generation	DALL-E, Midjourney
Code Generation	Code Llama
Text-to-Speech	Eleven Labs
Video Generation	Sora

2 Generative AI vs Traditional AI

2.1 Traditional AI (Pre-GenAI Era)

- **Focus:** Pattern recognition and relationship mapping between inputs and outputs
- **Method:** Learn from labeled data (input-output pairs)

- **Examples:** Classification (spam detection, cancer diagnosis), Regression (stock price prediction, weather forecasting)
- **Limitation:** Cannot generate new data; only predicts outputs

2.2 Generative AI

- **Focus:** Understanding the entire distribution/nature of data
- **Method:** Learns how data looks in the real world, then generates new samples from that distribution
- **Key Advantage:** Outputs feel human-created

3 Applications of Generative AI

3.1 Creative & Business Writing

- Blog writing, email drafting, content creation
- Integration with Gmail, Google Docs

3.2 Software Development

- Auto-completion tools
- Code debugging assistance
- Error resolution

3.3 Customer Support

- AI chatbots for large-scale user support
- Escalation to human agents when needed
- Used by: Ola, Uber, Zomato, Swiggy

3.4 Education

- Doubt clearing, personalized curricula
- Simplified explanations of complex topics

3.5 Designing

- Thumbnail generation, infographics
- Advertisement creation (using Sora, Runway)

4 Practical Scenario: HR Recruiter Case Study

Problem: You're an HR recruiter needing to hire a Backend Engineer with 2–4 years of experience.

4.1 Manual Hiring Steps

1. Draft Job Description (JD)
2. Post on job platforms (LinkedIn, Naukri)
3. Screen applications & shortlist candidates
4. Schedule and conduct interviews
5. Send offer letters
6. Onboard new employee

5 Evolution of AI Solutions for Hiring

5.1 Chatbot 1: Simple LLM-Based Chatbot

Capabilities:

- Draft JDs
- Suggest job platforms
- Provide generic screening advice
- Generate interview question banks
- Draft emails and offer letters

Problems:

- × Reactive: Needs human prompting at every step
- × No Memory: Forgets past context
- × Generic Advice: Not tailored to company specifics
- × No Actions: Cannot execute tasks independently (post jobs, send emails)

5.2 Chatbot 2: RAG-Based Chatbot (Retrieval-Augmented Generation)

Improvement: Connected chatbot to company knowledge base

Documents Fed:

- Past JD templates
- Hiring strategies/playbooks
- Salary bands
- Shortlisting criteria
- Interview question banks
- Offer letter templates
- Welcome email templates

- Employee policies

Advantage: Provides tailored/company-specific advice instead of generic suggestions

Still Problems:

- × Reactive behavior remains
- × No context awareness/memory
- ✓ Generic advice problem solved
- × Cannot perform actions independently

5.3 Chatbot 3: Tool-Augmented Chatbot

Improvement: Integrated the chatbot with various tools

Tools Integrated:

- LinkedIn API → Post jobs
- Resume Parser → Read and analyze PDF resumes
- Calendar API → Check availability, schedule interviews
- Email API → Send and receive emails
- HR Management Software → Automate onboarding

Advantages:

- Can draft AND send emails
- Can create AND post JDs
- Can schedule interviews
- Can trigger onboarding processes

Still Problems:

- × Reactive behavior
- × No context awareness/memory
- ✓ Generic advice solved
- ✓ Actions now possible
- × Cannot adapt autonomously

5.4 Chatbot 4: Agentic AI Chatbot

Improvement: Created a proactive, context-aware, and adaptable system

Key Capabilities:

- **Goal-Oriented:** Understands the end goal (hire a backend engineer)
- **Planning:** Creates execution plan autonomously

- **Reasoning:** Interprets, concludes, and decides
- **Context-Aware:** Remembers past steps and knows what to do next
- **Adaptive:** Identifies problems and proposes/executes solutions
- **Autonomous:** Takes initiative; minimal human involvement

Demonstrated Behaviors:

- Automatically drafts JD using company-specific documents
- Posts jobs to LinkedIn/Naukri via APIs
- Continuously monitors applications
- Self-identifies problem: “Only 2 applications received (below expectations)”
- Suggests solutions: Broaden JD to Full Stack, promote on LinkedIn
- Screens resumes automatically using parser tool
- Categorizes candidates: Strong (2), Partial (3), Weak (3)
- Schedules interviews based on calendar availability
- Sends reminders before interviews
- Tracks offer letter acceptance
- Triggers entire onboarding process automatically
- Sets up intro meetings

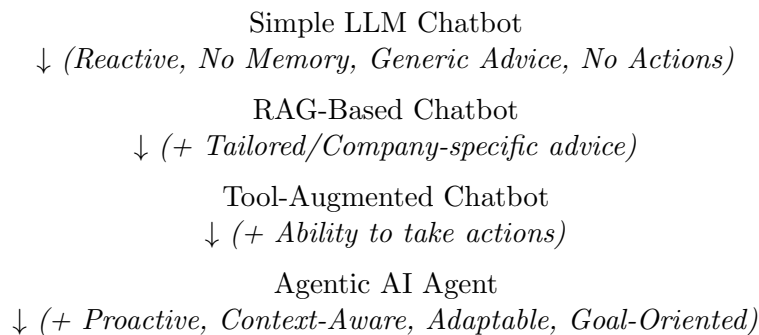
Human Role: Only provides approvals and monitors the process

6 Key Differences: Generative AI vs Agentic AI

Feature	Generative AI	Agentic AI
Primary Focus	Content Creation (text, images, code, video)	Goal Achievement (completing a task/objective)
End Goal	Generate content	Achieve a given goal through planning & execution
Behavior	Reactive: Responds to direct prompts; human guides every step	Proactive/Autonomous: Takes initiative; once given a goal, handles planning & execution
Context	Limited/no inherent context awareness/memory	Context-aware; remembers past steps and knows what to do next
Planning	Doesn't plan multi-step workflows autonomously	Plans entire workflows to achieve goals
Action	Provides textual replies; cannot perform external actions	Performs external actions by integrating with tools

Feature	Generative AI	Agentic AI
Adaptability	Doesn't self-identify problems or adapt strategies	Autonomously identifies problems and proposes/executes solutions
Relationship	A capability; building block/subset of Agentic AI	A behavior; broader concept using GenAI (LLMs) for planning/reasoning

7 Summary: The Evolution Journey



8 Final Key Takeaways

- GenAI = Content Creation Focus (text, image, video, code)
- Agentic AI = Goal Achievement Focus (plan + execute to complete objectives)
- GenAI is Reactive → Agentic AI is Proactive/Autonomous
- GenAI is a Building Block of Agentic AI
- Agentic AI uses GenAI (LLMs) for planning, reasoning, and tool orchestration

“Generative AI is a capability, whereas Agentic AI is a behavior.”

Term	Definition
Generative AI	AI that creates new content resembling human-created data
Agentic AI	AI that autonomously works toward achieving goals
RAG	Retrieval-Augmented Generation: GenAI + company-specific knowledge base
Tool-Augmented	Chatbot with ability to use external tools/APIs
LLM	Large Language Model (brain of the AI system)
Context Awareness	Ability to remember past interactions
Adaptability	Ability to modify plans when unexpected conditions arise
Proactive	Takes initiative without human prompting

Term	Definition
Reactive	Responds only when prompted by humans
Human-in-the-Loop	System includes human approvals at critical junctures